Supplementary Information for:
Identifiability of a point defect model for oxide growth on AISI 347

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S1 Model derivation

This section gives the complete derivation of the model used in the main text. The mechanistic framework is the Point Defect Model (PDM) formulation of Li et al. (2020) for oxide growth in supercritical water, specialized here from supercritical to subcritical high-temperature water and applied to the duplex oxide on Nb-stabilized X6CrNiNb18-10 (AISI 347) under the exposure conditions of Veile et al. (2024). Equation numbers of the form "Eq. (n) of the supercritical-water formulation" refer to that source; boxed (displayed and labeled) equations are the working forms used in this study.

S1.1 System, geometry, and species

The scale is duplex. Coordinates are taken with the origin at the barrier-layer/outer-layer (bl/ol) interface:

$$\underbrace{\text{metal (m)}}_{x=L_{\text{bl}}} \Big| \underbrace{\text{barrier layer (bl), MO}_{\chi/2}}_{x=0} \Big| \underbrace{\text{outer layer (ol), MO}_{\delta/2}}_{x=-L_{\text{ol}}} \Big| \text{high-temperature water (e).} \tag{1}$$

That is, the metal/barrier-layer (m/bl) interface sits at $x = L_{\text{bl}}$, the bl/ol interface at $x = 0$, and the ol/water interface at $x = -L_{\text{ol}}$.

The microstructural identification of the PDM objects for X6CrNiNb18-10 in boiling-water-reactor hydrothermal water, following the phase analysis of Veile et al. (2024), is given in Table S1.

Table S1: Identification of the PDM layers with the observed oxide microstructure on X6CrNiNb18-10.

PDM object	Observed layer	Phase (XPS/EDX)	Valences
barrier layer (bl)	inner nanocrystalline Cr-rich layer	$FeCr_2O_4$ / $NiCr_2O_4$ spinel	$\chi = 8/3$ (spinel-averaged cation)
outer layer (ol)	discrete Fe-rich crystals	Fe_3O_4 (magnetite), partly Fe_2O_3	$\delta = 8/3$, $\theta = 3$
— (inert)	Ni enrichment zone at bl/matrix interface	metallic/NiO traces	not modeled ($\mathrm{d}L_{\text{Ni}}/\mathrm{d}t \approx 0$)

The Kröger–Vink species in the barrier layer are: the cation vacancy $V_M^{\chi\prime}$, the cation interstitial $M_i^{\chi+}$, the oxygen vacancy $V_O^{\bullet\bullet}$, the oxide ion O_O, the lattice cation M_M, and the electron e'. Constants: $\gamma = F/(RT)$; Ω_{bl}, Ω_{ol} are the molar volumes per cation of the barrier-layer and outer-layer oxides; Ω_m is the molar volume of the metal; and $\text{PBR} = \Omega_{\text{bl}}/\Omega_m$ is the Pilling–Bedworth ratio.

The environment corresponds to the conditions of Veile et al. (2024):

- $T = 513.15$ K (240°C), $P = 7$ MPa, i.e. liquid water with density $\rho \approx 0.8137$ kg/L (IAPWS-97).
- Since $\rho \gg 0.1$ g/cm^3, the electrochemical-oxidation (high-density) branch of the supercritical-water PDM formulation applies: this is the classical condensed-water PDM limit of the model.
- Dissolved oxygen $[O_2] = 0.4$ ppm; ultrapure water with conductivity 0.055 μS/cm.
- Neutral pH at temperature: $\mathrm{p}K_w(240\,^\circ\mathrm{C}) \approx 11.2$, hence pH ≈ 5.6.
- Dimensionless relative volumetric concentrations (Eqs. (13) and (14) of the supercritical-water formulation), with reference concentration $C^0 = 1$ mol/L:

$$C_{\mathrm{H_2O}} = \frac{1000\,\rho}{18.015\,C^0_{\mathrm{H_2O}}} \approx 45.2, \tag{2}$$

$$C_{\mathrm{O}} = \frac{[\mathrm{O_2}]_{\mathrm{ppm}}\,\rho \times 10^{-3}}{32.00\,C^0_{\mathrm{O}}} \approx 1.02 \times 10^{-5}. \tag{3}$$

- Potential V: open circuit (no polarization applied in the autoclave). V is set by the O_2/H_2O redox couple (the electrochemical corrosion potential) and is constant over an exposure; it is therefore absorbed into effective rate constants (Section S1.6). The open-circuit closure $i_{\mathrm{pol}} = 0$ (Eq. (46) of the supercritical-water formulation) becomes relevant only if V is made explicit (parametric studies in the electrochemical corrosion potential).

S1.2 Interfacial reactions

The complete reaction set of the supercritical-water PDM formulation is retained, with the high-density/electrochemical-oxidation significance of each reaction carried over to subcritical hydrothermal water. Here χ and δ are the cation valences in the barrier and outer layers, and θ is the valence in the dissolved/further-oxidized product. The reactions at the three interfaces are listed in Tables S2–S4.

Table S2: Reactions at the m/bl interface ($x = L_{\mathrm{bl}}$).

#	Reaction	Role	Lattice
1	$m + V_M^{\chi'} \to M_M + v_m + \chi e'$	annihilates cation vacancy; cation feed for ol (vacancy path)	conservative
2	$m \to M_i^{\chi+} + v_m + \chi e'$	generates cation interstitial; cation feed for ol (interstitial path)	conservative
3	$m \to M_M + (\chi/2)\,V_O^{\bullet\bullet} + \chi e'$	generates oxygen vacancies: **bl growth into the metal**	**non-conservative**

Only Reaction 3 (growth) and Reactions 10/10′ (destruction) move the barrier-layer interfaces; only Reaction 11 destroys the outer layer. These four reactions define the thickness ordinary differential equations.

S1.3 Potential distribution

The postulates of the supercritical-water formulation (its Section 2.1) are: potential drops exist at the m/bl and bl/e interfaces; the field strength ε_f in the barrier layer is constant (buffered, independent of V and L_{bl}); and the bl/e drop is linear in V and pH and independent of L_{bl}:

$$\phi_{\mathrm{bl/e}} = \alpha V + \beta\,\mathrm{pH} + \phi^0, \quad \text{(Eq. (1) of the supercritical-water formulation)} \tag{4}$$

$$V = \phi_{\mathrm{m/bl}} + \varepsilon_f L_{\mathrm{bl}} + \phi_{\mathrm{bl/e}}, \quad \text{(Eq. (2) of the supercritical-water formulation)} \tag{5}$$

$$\phi_{\mathrm{m/bl}} = (1-\alpha)V - \varepsilon_f L_{\mathrm{bl}} - \beta\,\mathrm{pH} - \phi^0. \quad \text{(Eq. (3) of the supercritical-water formulation)} \tag{6}$$

Table S3: Reactions at the bl/ol interface ($x = 0$).

#	Reaction	Role	Lattice
4	$M_M + (\delta/4)\,O_2 + \chi e' \to V_M^{\chi'} + MO_{\delta/2}$	generates cation vacancy; deposits ol (direct O_2 route)	conservative
4′	$M_M \to V_M^{\chi'} + M^{\delta+} + (\delta-\chi)\,e'$	generates cation vacancy; cation ejected (dissolution/re-precipitation route; favored in condensed water)	conservative
5	$M_i^{\chi+} + (\delta/4)\,O_2 + \chi e' \to MO_{\delta/2}$	annihilates interstitial; deposits ol	conservative
5′	$M_i^{\chi+} \to M^{\delta+} + (\delta-\chi)\,e'$	annihilates interstitial; cation ejected	conservative
6	$V_O^{\bullet\bullet} + \frac{1}{2}O_2 + 2e' \to O_O$	annihilates oxygen vacancy (O_2 present)	conservative
7	$V_O^{\bullet\bullet} + H_2O + 2e' \to O_O + H_2$	annihilates oxygen vacancy (water route)	conservative
8	$V_O^{\bullet\bullet} + H_2O \to O_O + 2H^+$	annihilates oxygen vacancy, proton release (classical PDM; H^+ available in condensed water)	conservative
9	$H_2O + 2e' \to \frac{1}{2}H_2 + OH^-$	cathodic partial reaction	(no defect)
10	$MO_{\chi/2} + \frac{\delta-\chi}{4}\,O_2 \to MO_{\delta/2}$	converts bl to ol: **bl destruction** (chemical)	**non-conservative**
10′	$MO_{\chi/2} + \chi H^+ \to M^{\delta+} + (\chi/2)\,H_2O + (\delta-\chi)\,e'$	proton-assisted **bl dissolution**	**non-conservative**

Table S4: Reaction at the ol/water interface ($x = -L_{\mathrm{ol}}$).

#	Reaction	Role	Lattice
11	$MO_{\delta/2} + \frac{\theta-\delta}{4}\,O_2 \to MO_{\theta/2}(d)$	**ol destruction**: dissolution / further oxidation (magnetite → hematite, seen in the XPS of Veile et al.)	**non-conservative**

Here α is the polarizability of the bl/e interface ($\partial\phi_{\mathrm{bl/e}}/\partial V$); $\beta = \partial\phi_{\mathrm{bl/e}}/\partial\,\mathrm{pH}$ (< 0, typically -0.005 to -0.05 V); and ϕ^0 is the drop at $V = 0$, pH $= 0$, $L_{\mathrm{bl}} = 0$. The porous outer layer carries no potential drop (the cathodic reactions occur at the bl/ol interface via O_2/H_2O transport through pores; there is no remote cathode current in an autoclave at open circuit).

S1.4 Rate constants (activated-complex theory with partial charges)

Every interfacial rate constant takes the form (Eq. (8) of the supercritical-water formulation, with the derivation in Appendix A of Li et al. (2020)):

$$k_i = k_i^0\,\exp(a_i V)\,\exp(b_i L_{\mathrm{bl}})\,\exp(c_i\,\mathrm{pH}), \tag{7}$$

with the temperature dependence carried by the base standard rate constant (Eq. (9) of the supercritical-water formulation):

$$k_i^{00} = k_i^{00\prime}\,\exp\left[-\frac{\alpha_i\,\Delta G^0_{R_i}}{R}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right]. \tag{8}$$

The coefficients (Table 1 of the supercritical-water formulation), with α_i the transfer coefficient of Reaction i, are given in Table S5.

Table S5: Exponential coefficients of the interfacial rate constants, Eq. (7).

i	a_i (1/V)	b_i (1/cm)	c_i
1, 2, 3 (m/bl)	$\alpha_i\chi\gamma(1-\alpha)$	$\boldsymbol{-\alpha_i\chi\gamma\varepsilon_f}$	$-\alpha_i\chi\gamma\beta$
4, 5, 6, 7, 10, 11	0	0	0
4′, 5′	$\alpha_i\delta\gamma\alpha$	0	$\alpha_i\delta\gamma\beta$
8, 9	$2\alpha_i\gamma\alpha$	0	$2\alpha_i\gamma\beta$
10′	$\alpha_{10'}(\delta-\chi)\gamma\alpha$	0	$\alpha_{10'}(\delta-\chi)\gamma\beta$

The m/bl reactions inherit their V-, L_{bl}-, and pH-dependence from $\phi_{\mathrm{m/bl}}$; the bl/ol electrochemical reactions inherit theirs from $\phi_{\mathrm{bl/e}}$, hence $b_i = 0$ there. The single most important structural fact is

$$b_3 = -\alpha_3 \chi \gamma \varepsilon_f < 0 : \tag{9}$$

the growth reaction decelerates exponentially with film thickness. This is what generates direct-logarithmic/pseudo-parabolic kinetics and a finite steady-state thickness.

The steady-state defect flux balances (rates equal at both interfaces for each defect; Eqs. (10)–(12) of the supercritical-water formulation) are

$$k_1\, C_{V_M}^{L} = k_4 C_{\mathrm{O}}^{n} + k_{4'}, \tag{10}$$

$$k_2 = (k_5 C_{\mathrm{O}}^{n} + k_{5'})\, C_{M_i}^{0}, \tag{11}$$

$$\frac{\chi}{2}\, k_3 = \left(k_6 C_{\mathrm{O}}^{m} + k_7 C_{\mathrm{H_2O}}^{p} + k_8 C_{\mathrm{H_2O}}^{p}\right) C_{V_O}^{0}. \tag{12}$$

The kinetic orders are: n (Reactions 4 and 5 in C_{O}), m (Reaction 6 in C_{O}), p (Reactions 7–9 in $C_{\mathrm{H_2O}}$), q (Reaction 10 in C_{O}), r (Reaction 11 in C_{O}), and s (Reaction $10'$ in the H^+ activity ratio). Li et al. (2020) assume $n = m = q = r = 1/2$.

S1.5 Thickness evolution equations (the governing ODEs)

S1.5.1 Barrier layer

Only the non-conservative reactions contribute (Eq. (16) of the supercritical-water formulation, with porosity correction $\rho_{\mathrm{bl}}^{0}/\rho_{\mathrm{bl}}$):

$$\frac{\mathrm{d}L_{\mathrm{bl}}}{\mathrm{d}t} = \Omega_{\mathrm{bl}} \left[k_3 - k_{10} C_{\mathrm{O}}^{q} - k_{10'} \left(C_{\mathrm{H^+}}/C_{\mathrm{H^+}}^{0}\right)^{s}\right] \frac{\rho_{\mathrm{bl}}^{0}}{\rho_{\mathrm{bl}}}. \tag{13}$$

Substituting $k_3 = k_3^0 \exp(a_3 V) \exp(b_3 L_{\mathrm{bl}}) \exp(c_3\,\mathrm{pH})$ at constant V and pH (Eqs. (21)–(23) of the supercritical-water formulation) gives the working form

$$\boxed{\frac{\mathrm{d}L_{\mathrm{bl}}}{\mathrm{d}t} = A_{\mathrm{bl}}\, \exp(b_3 L_{\mathrm{bl}}) - C_{\mathrm{bl}}} \tag{14}$$

with

$$A_{\mathrm{bl}} = \Omega_{\mathrm{bl}}\, k_3^0\, \exp(a_3 V)\, \exp(c_3\,\mathrm{pH})\, \frac{\rho_{\mathrm{bl}}^{0}}{\rho_{\mathrm{bl}}}, \qquad \text{(Eq. (22) of the supercritical-water formulation)} \tag{15}$$

$$C_{\mathrm{bl}} = \Omega_{\mathrm{bl}} \left[k_{10} C_{\mathrm{O}}^{q} + k_{10'} \left(C_{\mathrm{H^+}}/C_{\mathrm{H^+}}^{0}\right)^{s}\right] \frac{\rho_{\mathrm{bl}}^{0}}{\rho_{\mathrm{bl}}}. \qquad \text{(Eq. (23) of the supercritical-water formulation)} \tag{16}$$

(The printed “k_9” in Eq. (23) of the supercritical-water formulation is a typographical error for k_{10}, as required by consistency with its Eqs. (16), (17), and (44); see Section S1.10.)

The **closed-form solution** for constant A_{bl}, b_3, C_{bl} and initial condition $L_{\mathrm{bl}}(0) = L_0$ (Eqs. (24) and (39) of the supercritical-water formulation) is

$$L_{\mathrm{bl}}(t) = L_0 - \frac{1}{b_3} \ln\left[1 + \frac{A_{\mathrm{bl}}}{C_{\mathrm{bl}}}\, e^{b_3 L_0} \left(e^{-b_3 C_{\mathrm{bl}} t} - 1\right)\right] - C_{\mathrm{bl}}\, t. \tag{17}$$

The **steady state** ($\mathrm{d}L_{\mathrm{bl}}/\mathrm{d}t = 0$; Eqs. (17)–(20) of the supercritical-water formulation) is

$$L_{\mathrm{bl,ss}} = \frac{1-\alpha}{\varepsilon_f}\,V - \frac{\beta}{\varepsilon_f}\,\mathrm{pH} - \frac{1}{\alpha_3\chi\varepsilon_f\gamma}\ln\left[\frac{k_{10}C_{\mathrm{O}}^{q} + k_{10'}\left(C_{\mathrm{H^+}}/C_{\mathrm{H^+}}^{0}\right)^{s}}{k_3^0}\right]. \tag{18}$$

$L_{\mathrm{bl,ss}}$ is linear in V and pH, the PDM signature. In ultrapure hydrothermal water (0.055 μS/cm, pH $\approx$ 5.6, $C_{\mathrm{O}} \approx 10^{-5}$) the destruction terms are very small: $C_{\mathrm{bl}} \to 0$, the steady state moves to very large L, and over a finite window

$$L_{\mathrm{bl}}(t) \approx L_0 - \frac{1}{b_3}\ln\Big(1 - b_3 A_{\mathrm{bl}}\, e^{b_3 L_0}\, t\Big) \tag{19}$$

— direct logarithmic growth, which over 72–480 h is numerically indistinguishable from the parabolic $t^{0.5}$ law fitted by Veile et al. (their Cr layer: $n = 0.4964$, no saturation at 480 h). Quantifying this equivalence is one of the results of the study.

The limiting-case map (what the data can select between) is:

- $C_{\mathrm{bl}} = 0$, $|b_3|L \ll 1$: linear growth, $L \approx L_0 + A_{\mathrm{bl}}t$.
- $C_{\mathrm{bl}} = 0$, $|b_3|L = O(1)$: direct-logarithmic (pseudo-parabolic over a decade of t).
- $C_{\mathrm{bl}} > 0$: logarithmic $\to$ saturation at $L_{\mathrm{bl,ss}}$.

S1.5.2 Outer layer and the constant-volume constraint

The outer layer grows from cations transmitted through the barrier layer (fluxes $k_1 C_{V_M}^{L}$ via vacancies and k_2 via interstitials, deposited by Reactions 4/4′/5/5′), plus bl→ol conversion (Reactions 10/10′), minus dissolution (Reaction 11) (Eq. (26) of the supercritical-water formulation):

$$\frac{\mathrm{d}L_{\mathrm{ol}}}{\mathrm{d}t} = \Omega_{\mathrm{ol}}\left[k_1 C_{V_M}^{L} + k_2 + k_{10}C_{\mathrm{O}}^{q} + k_{10'}\left(C_{\mathrm{H^+}}/C_{\mathrm{H^+}}^{0}\right)^{s} - k_{11}C_{\mathrm{O}}^{r}\right]\frac{\rho_{\mathrm{ol}}^{0}}{\rho_{\mathrm{ol}}}. \tag{20}$$

Constant-volume growth is imposed (marker experiments show that the bl/ol interface stays at the original metal surface): the total metal consumption rate equals the rate of barrier-layer advance into the metal (Eqs. (29)–(32) of the supercritical-water formulation):

$$\frac{\mathrm{d}L_m}{\mathrm{d}t} = \Omega_m\left(k_1 C_{V_M}^{L} + k_2 + k_3\right) \qquad \text{(metal destruction, Reactions 1+2+3)}, \tag{21}$$

$$\frac{\mathrm{d}L_{\mathrm{bl}}}{\mathrm{d}t} = \Omega_{\mathrm{bl}}\left(k_3 - k_d\right), \qquad k_d := k_{10}C_{\mathrm{O}}^{q} + k_{10'}\left(C_{\mathrm{H^+}}/C_{\mathrm{H^+}}^{0}\right)^{s}, \tag{22}$$

where the metal, being dense, carries no porosity correction and the barrier-layer rate is written with $\rho_{\mathrm{bl}}^{0}/\rho_{\mathrm{bl}} = 1$; setting $\mathrm{d}L_m/\mathrm{d}t = \mathrm{d}L_{\mathrm{bl}}/\mathrm{d}t$ gives

$$\boxed{k_1 C_{V_M}^{L} + k_2 + k_3 = \mathrm{PBR}\left(k_3 - k_d\right)} \qquad \text{(Eq. (32) of the supercritical-water formulation).} \tag{23}$$

For an n-type barrier layer (interstitials/oxygen vacancies dominate, $k_1 C_{V_M}^{L} \ll k_2$) and $k_d \approx 0$ (Eq. (33) of the supercritical-water formulation):

$$k_2 = (\mathrm{PBR} - 1)\,k_3. \tag{24}$$

Substituting Eq. (23) into the outer-layer equation (20), the k_d terms combine exactly, $(\mathrm{PBR}-1)k_3 - \mathrm{PBR}\,k_d + k_d = (\mathrm{PBR}-1)(k_3 - k_d)$, giving (Eq. (41) of the supercritical-water formulation)

$$\boxed{\begin{aligned}\frac{\mathrm{d}L_{\mathrm{ol}}}{\mathrm{d}t} &= \Omega_{\mathrm{ol}}\left[(\mathrm{PBR}-1)(k_3-k_d) - k_{11}C_{\mathrm{O}}^{r}\right] \\ &= (\mathrm{PBR}-1)\frac{\Omega_{\mathrm{ol}}}{\Omega_{\mathrm{bl}}}\frac{\mathrm{d}L_{\mathrm{bl}}}{\mathrm{d}t} - \Omega_{\mathrm{ol}}\,k_{11}C_{\mathrm{O}}^{r}\,\frac{\rho_{\mathrm{ol}}^{0}}{\rho_{\mathrm{ol}}}\end{aligned}} \tag{25}$$

Retaining both porosity corrections instead would give the coefficient $(\Omega_{\mathrm{ol}}/\Omega_{\mathrm{bl}})(\rho_{\mathrm{ol}}^{0}/\rho_{\mathrm{ol}})(\mathrm{PBR} - \rho_{\mathrm{bl}}/\rho_{\mathrm{bl}}^{0})$, so the dense-oxide convention, and not merely equal porosity between the layers, is what reduces it to the form above. The convention affects the physical quantities recovered from the fit, not the fitted parameters themselves. With negligible barrier-layer destruction the thickness ratio is then fixed by stoichiometry alone (Eqs. (36) and (37) of the supercritical-water formulation):

$$\frac{\mathrm{d}L_{\mathrm{ol}}}{\mathrm{d}L_{\mathrm{bl}}} = (\mathrm{PBR}-1)\,\frac{\Omega_{\mathrm{ol}}}{\Omega_{\mathrm{bl}}} =: \mathrm{PBR}_{\mathrm{eff}}. \tag{26}$$

Closed form. Li et al. (2020) write it (their Eqs. (42)–(45)) with $A_{\mathrm{ol}} = (\mathrm{PBR}-1)(\Omega_{\mathrm{ol}}/\Omega_{\mathrm{bl}})A_{\mathrm{bl}}$ and $C_{\mathrm{ol}} = \Omega_{\mathrm{ol}}[(\mathrm{PBR}-1)k_d + k_{11}C_{\mathrm{O}}^{r}]$ (so that $\mathrm{d}L_{\mathrm{ol}}/\mathrm{d}t = A_{\mathrm{ol}}\exp(b_3 L_{\mathrm{bl}}) - C_{\mathrm{ol}}$):

$$L_{\mathrm{ol}}(t) = \frac{A_{\mathrm{ol}}}{A_{\mathrm{bl}}b_3}\ln\Big[A_{\mathrm{bl}}\,e^{b_3 L_{\mathrm{bl}}(t)} - C_{\mathrm{bl}}\Big] - C_{\mathrm{ol}}\,t + D_{\mathrm{ol}}, \tag{27}$$

$$D_{\mathrm{ol}} = -\frac{A_{\mathrm{ol}}}{A_{\mathrm{bl}}b_3}\ln\Big[A_{\mathrm{bl}}\,e^{b_3 L_0} - C_{\mathrm{bl}}\Big] + L_{\mathrm{ol}}(0). \tag{28}$$

Because $\mathrm{d}L_{\mathrm{ol}}/\mathrm{d}t$ is an exact affine function of $\mathrm{d}L_{\mathrm{bl}}/\mathrm{d}t$, the reduced system also integrates trivially:

$$\begin{gathered} L_{\mathrm{ol}}(t) = \mathrm{PBR}_{\mathrm{eff}}\,(L_{\mathrm{bl}}(t) - L_0) + C_x\,t + L_{\mathrm{ol}}(0), \\ \mathrm{PBR}_{\mathrm{eff}} := (\mathrm{PBR}-1)\frac{\Omega_{\mathrm{ol}}}{\Omega_{\mathrm{bl}}}, \qquad C_x := -\Omega_{\mathrm{ol}}\,k_{11}C_{\mathrm{O}}^{r}\ (\le 0)\end{gathered} \tag{29}$$

— an independent verification target alongside Eq. (27).

The total scale thickness is $L_{\mathrm{tot}} = L_{\mathrm{bl}} + L_{\mathrm{ol}}$. The weight gain (not measurable for the cube specimens of Veile et al., kept for completeness; Eq. (28) of the supercritical-water formulation) is $\mathrm{d}(\Delta w)/\mathrm{d}t = r_{\mathrm{bl}}\rho_{\mathrm{bl}}\,\mathrm{d}L_{\mathrm{bl}}/\mathrm{d}t + r_{\mathrm{ol}}\rho_{\mathrm{ol}}\,\mathrm{d}L_{\mathrm{ol}}/\mathrm{d}t$, with r the oxygen mass fraction of each oxide; a closed seven-parameter form is given in Eqs. (48)–(56) of the supercritical-water formulation.

S1.5.3 Interfacial current (for extensions with explicit corrosion potential)

The partial currents (Appendix B of Li et al. (2020)) are $i_1 = \chi F k_1 C_{V_M}^{L}$, $i_2 = \chi F k_2$, $i_3 = \chi F k_3$, $i_4 = -\chi F k_4 C_{\mathrm{O}}^{n}$, $i_5 = -\chi F k_5 C_{\mathrm{O}}^{n} C_{M_i}^{0}$, $i_6 = -2Fk_6 C_{\mathrm{O}}^{m} C_{V_O}^{0}$, $i_7 = -2Fk_7 C_{\mathrm{H_2O}}^{p} C_{V_O}^{0}$, $i_9 = -2Fk_9 C_{\mathrm{H_2O}}^{p}$, $i_{4'} = (\delta - \chi)Fk_{4'}$, etc. At open circuit the sum of the partial currents vanishes (Eq. (46) of the supercritical-water formulation), which fixes $V = V_{\mathrm{ocp}}$ given the k_i^{0}.

S1.6 Reduction to the identifiable apparent-parameter model

At open circuit, V and pH are constant over an exposure, so $\exp(a_3 V + c_3\,\mathrm{pH})$ is a constant multiplier: it cannot be separated from k_3^{0} by thickness-versus-time data alone. The observable model is therefore the **apparent-parameter system** (identical in structure to Table 5 of Li et al. (2020)):

$$\frac{\mathrm{d}L_{\mathrm{bl}}}{\mathrm{d}t} = A_{\mathrm{bl}}\,\exp(b_3 L_{\mathrm{bl}}) - C_{\mathrm{bl}}, \qquad L_{\mathrm{bl}}(0) = L_0, \tag{30}$$

$$\frac{\mathrm{d}L_{\mathrm{ol}}}{\mathrm{d}t} = \mathrm{PBR}_{\mathrm{eff}}\,(A_{\mathrm{bl}}\,\exp(b_3 L_{\mathrm{bl}}) - C_{\mathrm{bl}}) + C_x, \qquad L_{\mathrm{ol}}(0) = L_{\mathrm{ol},0}, \tag{31}$$

with $C_x := -\Omega_{\mathrm{ol}} k_{11} C_{\mathrm{O}}^r \le 0$ (outer-layer dissolution / further oxidation only).

The parameter vector is

$$\theta = \big\{A_{\mathrm{bl}}\ [\mathrm{nm/h}],\ b_3\ [1/\mathrm{nm}]\ (<0),\ C_{\mathrm{bl}}\ [\mathrm{nm/h}]\ (\ge 0),\ \mathrm{PBR}_{\mathrm{eff}}\ [-]\ (>0),\ C_x\ [\mathrm{nm/h}]\ (\le 0),\ L_0\ [\mathrm{nm}]\ (>0),\ L_{\mathrm{ol},0}\ [\mathrm{nm}]\ (\ge 0)\big\} \tag{32}$$

— 7 parameters, 2 ODEs, data at 3 times for 2 layers (plus a Ni consistency check). The accepted model M4 fixes $C_{\mathrm{bl}} = C_x = 0$ and fits the remaining five; the initial outer-layer thickness $L_{\mathrm{ol},0}$ is the parameter whose addition resolves the degenerate ridge of M3 (main text, Section 3.1).

Physical quantities are recovered after fitting via

$$\varepsilon_f = \frac{-b_3}{\alpha_3 \chi \gamma} \qquad [\text{from } b_3 = -\alpha_3 \chi \gamma \varepsilon_f], \tag{33}$$

$$k_3^0 = \frac{A_{\mathrm{bl}}}{\Omega_{\mathrm{bl}}\,\exp(a_3 V + c_3\,\mathrm{pH})} \qquad [\text{order of magnitude, given } \alpha_3, V], \tag{34}$$

$$\mathrm{PBR} = 1 + \mathrm{PBR}_{\mathrm{eff}}\,\frac{\Omega_{\mathrm{bl}}}{\Omega_{\mathrm{ol}}}, \tag{35}$$

$$k_d = \frac{C_{\mathrm{bl}}}{\Omega_{\mathrm{bl}}}, \tag{36}$$

with $\alpha_3 \in [0.1, 0.5]$ (Li et al. (2020) fit $\alpha_3 = 0.12$ for HCM12A) and $\chi = 8/3$, $\gamma = F/RT = 22.6\ \mathrm{V}^{-1}$ at 513.15 K. Molar volumes per cation: $\Omega_{\mathrm{bl}} = 46.7/3 = 15.6\ \mathrm{cm}^3/\mathrm{mol}$ for $FeCr_2O_4$ (3 cations); $\Omega_{\mathrm{ol}} = 45.0/3 = 15.0\ \mathrm{cm}^3/\mathrm{mol}$ for Fe_3O_4; $\Omega_m \approx 7.1\ \mathrm{cm}^3/\mathrm{mol}$ for the fcc stainless steel. These are broadly consistent, as an independent cross-check rather than a re-derivation, with the theoretical PBR = 2.05 ($FeCr_2O_4$) and 2.10 (Fe_3O_4) reported in Table 3 of Li et al. (2020) for the same phases against their own representative steel molar volume; the fit itself recovers $\mathrm{PBR}_{\mathrm{eff}} = 1.096$, the apparent outer-to-barrier growth-rate ratio identified from the thickness data (not $\Omega_{\mathrm{bl}}/\Omega_m$ directly), related to the physical PBR through the equation above.

Priors/constraints (3 time points cannot determine 7 parameters freely). The only prior applied in every fit is the Gaussian log-space prior of Eq. (16) of the main text: $\ln L_0 \sim \mathcal{N}(\ln 2\ \mathrm{nm}, 0.60^2)$, whose $\pm 2\sigma$ range is 0.6–6.6 nm, matching the air-formed-film range below.

Earlier versions of this analysis also applied $\ln \mathrm{PBR}_{\mathrm{eff}} \sim \mathcal{N}(\ln 1.05, 0.20^2)$, on the grounds that 1.05 was the stoichiometric value. It is not: the constant-volume closure giving 1.05 requires the barrier layer to draw ≈ 30 at% Cr from an 18.8 at% alloy, and a chromium mass balance with an $FeCr_2O_4$ barrier gives 2.05 when all non-Ni iron reports to the outer layer. That prior has been removed; $\mathrm{PBR}_{\mathrm{eff}}$ is now determined by the likelihood alone (1.096 for M4). The apparent gap between 1.096 and 2.05 is examined in the main text and does not survive: the mass-balance figure is phase-dependent and spans $[0.52, 3.79]$ over the candidate barrier spinels, and once an outer-layer loss term is admitted the $FeCr_2O_4$ value is reached at no cost in χ^2. The physical ranges below motivate the remaining prior:

- L_0: air-formed film on high-Cr austenitic stainless steel, 1–5 nm (Li et al. (2020): 1.01 nm for 316L).
- $\mathrm{PBR_{eff}}$: no prior; fitted freely. The chromium mass balance admits $[0.52, 3.79]$ depending on the barrier phase assignment; the likelihood returns 1.096, inside that range.
- C_{bl}: ≥ 0, expected ≈ 0 (ultrapure water); both $C_{\mathrm{bl}} = 0$ (five-parameter model) and free C_{bl} are tested.
- b_3: strictly < 0. $|b_3| \sim \alpha_3\chi\gamma\varepsilon_f$; with $\varepsilon_f \in [10^2, 10^6]$ V/cm this spans $|b_3| \in [\sim 7 \times 10^{-5}, \sim 0.7]$ $\mathrm{nm^{-1}}$: the data must localize it.
- C_x: ≤ 0 by construction, and small.

The units convention for the implementation is thickness in nm and time in h. Conversions from the cgs units (cm, s) of Li et al. (2020): 1 cm/s $= 3.6 \times 10^{10}$ nm/h; 1 $\mathrm{cm^{-1}} = 10^{-7}$ $\mathrm{nm^{-1}}$. Sign conventions respected throughout: $b_3 < 0$, $C_{\mathrm{bl}} \geq 0$, $\beta < 0$, $D_{\mathrm{ol}} < 0$.

S1.7 Numerically stable evaluation of the closed forms

Equation (17) is not evaluated as written. For $C_{\mathrm{bl}} \to 0$ the ratio $A_{\mathrm{bl}}/C_{\mathrm{bl}}$ diverges while the bracket it multiplies vanishes, and in the fitted regime (C_{bl} driven to zero by the data) a direct evaluation loses all significance. The implementation therefore takes the $C_{\mathrm{bl}} = 0$ branch analytically,

$$L_{\mathrm{bl}}(t) = L_0 - \frac{1}{b_3}\,\mathrm{log1p}\Big(-b_3 A_{\mathrm{bl}} e^{\,b_3 L_0}\,t\Big)\,, \tag{37}$$

and for $C_{\mathrm{bl}} > 0$ forms the logarithm of the bracket in log space. Writing $z = -b_3 C_{\mathrm{bl}} t \geq 0$ and $u = 1 + (A_{\mathrm{bl}}/C_{\mathrm{bl}}) e^{b_3 L_0}(e^z - 1)$, the identity $e^z - 1 = e^z(1 - e^{-z})$ gives

$$\ln(u-1) = \ln A_{\mathrm{bl}} - \ln C_{\mathrm{bl}} + b_3 L_0 + z + \mathrm{log1p}\big(-e^{-z}\big)\,, \qquad \ln u = \mathrm{logaddexp}(0, \ln(u-1)), \tag{38}$$

in which every term is evaluated by a library routine that is exact in its own small-argument limit. The result is accurate in both the $|b_3|L \ll 1$ and the $|b_3|L = O(1)$ regimes, which is what the parity check of Section S1.9 confirms and what makes a 1000-member bootstrap over the closed form affordable.

S1.8 Nondimensionalization for the neural spectral-element solver

The characteristic scales are $t_c = 480$ h (the exposure window) and $L_c = 100$ nm (the observed Cr-layer scale). With $\tau = t/t_c$ and $\lambda = L/L_c$:

$$\frac{\mathrm{d}\lambda_{\mathrm{bl}}}{\mathrm{d}\tau} = \hat{A}\,\exp(\hat{b}\,\lambda_{\mathrm{bl}}) - \hat{C}, \qquad \lambda_{\mathrm{bl}}(0) = \lambda_0, \tag{39}$$

$$\frac{\mathrm{d}\lambda_{\mathrm{ol}}}{\mathrm{d}\tau} = \mathrm{PBR_{eff}}\left(\hat{A}\,\exp(\hat{b}\,\lambda_{\mathrm{bl}}) - \hat{C}\right) + \hat{C}_x, \tag{40}$$

with

$$\hat{A} = \frac{A_{\mathrm{bl}}\,t_c}{L_c}, \qquad \hat{b} = b_3 L_c, \qquad \hat{C} = \frac{C_{\mathrm{bl}}\,t_c}{L_c}, \qquad \hat{C}_x = \frac{C_x\,t_c}{L_c}, \qquad \lambda_0 = \frac{L_0}{L_c}. \tag{41}$$

For the expected regime (L_{bl}: $37 \to 134$ nm over $72 \to 480$ h with near-parabolic shape) these groups are $O(0.1$–$10)$: e.g. a pure-logarithmic fit gives $\hat{b} \approx -2 \ldots -4$, $\hat{A} \approx O(10)$, $\hat{C} \approx 0$. All learnable parameters are $O(1)$; a badly scaled parameter is an unlearnable parameter, so this scaling is applied from the start.

The residuals are enforced in Crank–Nicolson integral form on the Gauss–Lobatto–Legendre time grid (otherwise the conditioning of the spectral first-derivative matrix is the dominant error floor):

$$R_j = \lambda_j - \lambda_{j-1} - \frac{\Delta t_j}{2}\left(f(\lambda_{j-1}) + f(\lambda_j)\right), \qquad j = 1, \ldots, N_t - 1, \tag{42}$$

plus the initial-condition residuals $\lambda_{\mathrm{bl}}(0) - \lambda_0$ and $\lambda_{\mathrm{ol}}(0)$.

S1.9 Verification targets

Four verification targets follow from the derivation (results in Sections S5 and S6 below):

1. **Closed-form parity**: Radau integration of the apparent-parameter system of Section S1.6 must match Eqs. (17) and (27)–(29) to at least 10 significant digits for generic parameter sets ($A_{\mathrm{bl}}, b_3, C_{\mathrm{bl}}$, $\mathrm{PBR}_{\mathrm{eff}}, C_x, L_0$).

2. **Regression against Table 5 of Li et al. (2020)**: with the HCM12A values ($L_0 = 4.98 \times 10^{-5}$ cm, $b_3 = -6.08 \times 10^2$ cm^{-1}, $A_{\mathrm{bl}} = 3.15 \times 10^{-10}$ cm/s, $C_{\mathrm{bl}} = 3.66 \times 10^{-12}$ cm/s, $A_{\mathrm{ol}} = 3.66 \times 10^{-10}$ cm/s, $C_{\mathrm{ol}} = 4.25 \times 10^{-12}$ cm/s, $D_{\mathrm{ol}} = -4.18 \times 10^{-2}$) and the corresponding 316L values, reproduce the published thickness-versus-time behavior.

3. **Power-law refit**: least squares of $L = kt^n$ on the digitized data of Veile et al. (2024) must recover Cr ($k = 6.521$ nm, $n = 0.4964$) and Fe ($k = 64.51$ nm, $n = 0.2209$) within digitization error.

4. **Ni consistency**: the fitted model implies no Ni-layer dynamics; the data standard-deviation band (31–43 nm, approximately constant) must contain the assumed-constant line.

S1.10 Known errata in the sources

Documented here so they are not propagated:

- Li et al. (2020), Eq. (23): prints k_9 where k_{10} is required (consistency with Eqs. (16), (17), and (44) of that work).
- Veile et al. (2024): the exposure times are 72/168/480 h ("78 h" and "178 h" in the Results are typographical errors); the legend of their Fig. 9 swaps the n values between Fe and Cr, and the body text is authoritative (Fe: $n = 0.2209$; Cr: $n = 0.4964$).

S2 Spatial defect-transport model

This section gives the complete governing equations, boundary conditions, and initial condition for the spatial extension of Section 3.4 of the main text, which resolves oxygen- and cation-vacancy transport across the barrier layer under the field strength identified in Section S1. Nothing here is fit to the spatial data; every kinetic quantity entering this section (the growth prefactor, field exponent, and barrier-layer thickness $L(t)$) is frozen at its zero-dimensional value.

S2.1 Species, flux, and nondimensionalization

Two defect species are resolved on the barrier layer, $x \in [0, L]$ with the origin at the metal/barrier-layer interface: oxygen vacancies $c_{\mathrm{OV}}(x,t)$ (charge $z = +2$, generated at the metal/barrier-layer interface by the growth reaction and annihilated at the barrier/outer-layer interface, migrating outward) and cation vacancies $c_{\mathrm{CV}}(x,t)$ (charge $z = -\chi$ with $\chi = 8/3$ the spinel-averaged cation valence, generated at the barrier/outer-layer interface and annihilated at the metal/barrier-layer interface, migrating inward). Each species obeys a Nernst–Planck flux under the constant field ε_f identified in the zero-dimensional fit, with Einstein mobility $u = D\gamma$, $\gamma = F/(RT)$:

$$J = -D\,\frac{\partial c}{\partial x} + z\gamma\varepsilon_f D\,c. \tag{43}$$

Nondimensionalizing depth as $\xi = x/L$, concentration as $\hat{c} = c/c_0$ with the migration scale $c_0 = |J|/|v|$, $v = z\gamma\varepsilon_f D$ (the migration velocity), and flux as $\hat{J} = J/(|v|c_0) = \mathrm{sign}(v)$, the steady equation becomes

$$\frac{d}{d\xi}\left[-\frac{1}{\mathrm{Pe}}\frac{d\hat{c}}{d\xi} + s\hat{c}\right] = 0, \qquad \mathrm{Pe} = |z|\gamma\varepsilon_f L, \qquad s = \mathrm{sign}(z). \tag{44}$$

As discussed in the main text, the diffusivity D cancels from the Péclet number Pe: the nondimensional steady profile shape depends only on quantities the zero-dimensional fit already fixes, and D enters only the dimensional concentration scale c_0 (invisible to a composition measurement such as EDX).

S2.2 Boundary conditions

Both species obey a Robin (interfacial-kinetics) boundary condition at their annihilation interface, with κ the ratio of the interfacial rate constant to the migration velocity:

$$\text{OV (annihilation at } \xi = 1\text{):} \quad J = k_a\,c(L) \implies \hat{c}(1) = 1/\kappa_a, \tag{45}$$

$$\text{CV (annihilation at } \xi = 0\text{):} \quad |J| = k_1\,c(0) \implies \hat{c}(0) = 1/\kappa_1. \tag{46}$$

The generation flux at the opposite interface is anchored to the zero-dimensional growth kinetics rather than independently fit: the oxygen-vacancy generation flux is $J_{\mathrm{OV}} = (\chi/2)\,k_3(L)$ with $k_3(L)$ the barrier-layer growth rate constant at thickness L ($\mathrm{mol\,cm^{-2}\,s^{-1}}$, converted from the fitted dL_{bl}/dt), and the cation-vacancy generation flux is anchored to it by an assumed ratio, $|J_{\mathrm{CV}}| = r_{\mathrm{CV}} J_{\mathrm{OV}}$.

The parameters κ_a, κ_1, r_{CV}, and the two diffusivities D_{OV}, D_{CV} are *assumed*, not identified: per the synthetic identifiability gate of Section 3.4 of the main text, three exposure times cannot constrain interfacial transport rate constants, so these are treated as forward-only inputs, documented here for reproducibility rather than reported as findings. The values used throughout are $\varepsilon_f = 1.73 \times 10^4$ V/cm (the zero-dimensional fit), $\Omega_{\mathrm{bl}} = 15.6$ cm^3/mol, $D_{\mathrm{OV}} = 10^{-15}$ cm^2/s, $D_{\mathrm{CV}} = 10^{-16}$ cm^2/s, $\kappa_a = \kappa_1 = 1.25$, and $r_{\mathrm{CV}} = 0.5$.

S2.3 Analytic steady solution

Because the nondimensional flux is constant in steady state ($\hat{J} = s$ everywhere), Eq. (44) admits an exact closed-form solution: a uniform plateau at $\hat{c} = 1$ (the particular solution) plus a homogeneous exponential mode that satisfies the Robin condition at the annihilation boundary $\xi_{\mathrm{bc}} \in \{0, 1\}$:

$$\hat{c}(\xi) = 1 + (\hat{c}_{\mathrm{bc}} - 1)\exp\big[s\,\mathrm{Pe}\,(\xi - \xi_{\mathrm{bc}})\big]. \tag{47}$$

This is the reference solution the Newton and NSEM steady solves are verified against (Section S5).

S2.4 Transient (moving-boundary) problem

The transient problem is solved on a Landau-transformed fixed domain $\xi \in [0,1]$ that tracks the growing barrier-layer thickness $L(t)$, with $\tau = t/t_c$ the nondimensional time ($t_c = 480$ h, the exposure window). Writing $\mathrm{Pe}(\tau) = |z|\gamma\varepsilon_f L(\tau)$, $\hat{\jmath}(\tau) = k_3(L(\tau))/k_3(L_{\mathrm{ref}})$ the growth-flux ratio to a reference thickness $L_{\mathrm{ref}} = L(\tau{=}1)$, $\mathrm{adv}(\tau) = d\ln L/d\tau$ the Landau advection strength, and $\mathrm{mig}(\tau) = t_c/\tau_{\mathrm{mig}}(L(\tau))$ the flux-divergence prefactor ($\tau_{\mathrm{mig}} = L/|v|$ the migration transit time), the governing equation is

$$\frac{\partial \hat{c}}{\partial \tau} = \xi\,\mathrm{adv}(\tau)\,\frac{\partial \hat{c}}{\partial \xi} - \mathrm{mig}(\tau)\,\frac{\partial \hat{J}}{\partial \xi}, \qquad \hat{J} = -\frac{1}{\mathrm{Pe}(\tau)}\frac{\partial \hat{c}}{\partial \xi} + s\hat{c}. \tag{48}$$

Boundary conditions (per time node): at the generation boundary, $\hat{J} = s\,\hat{\jmath}(\tau)$ (the zero-dimensional growth-flux anchor); at the annihilation boundary, $\hat{J} = s\,\kappa\,\hat{c}$ (the same Robin closure as the steady problem). **Initial condition**: the instantaneous quasi-steady profile at τ_0, $\hat{c}(\xi,\tau_0) = \hat{\jmath}(\tau_0)\,\hat{c}_{\mathrm{ss}}(\xi;\mathrm{Pe}(\tau_0))$, with $\hat{c}_{\mathrm{ss}}$ the analytic steady shape of Eq. (47). Quasi-steadiness is quantified in Section S6 by comparing the transient solution at $\tau = 1$ against the instantaneous quasi-steady profile at that same moment.

S2.5 Synthetic-EDX instrument and heterogeneity model for the identifiability gate

The synthetic identifiability gate of the main text (20 replicate experiments per candidate parameter set) generates each replicate with the following instrument and heterogeneity model, anchored to the scan-to-scan scatter of Veile et al.'s replicate line scans:

- **Spatial sampling:** composition profiles sampled every 2 nm over $x \in [-120, 220]$ nm (170 points per scan), four element traces (O, Cr, Fe, Ni) per scan, three scans per replicate experiment.
- **EDX spatial resolution:** Gaussian smoothing with $\sigma = 2/2.355 \approx 0.85$ nm (2 nm full width at half maximum).
- **Per-scan position jitter:** each scan is rigidly shifted by a Gaussian offset with $\sigma = 0.945$ nm.
- **Compositional noise:** independent Gaussian noise per point, with standard deviation $\max(0.02\,|v|,\ 0.2)$ weight percent, that is 2% relative, floored at 0.2 absolute weight percent.
- **Scan-to-scan physical heterogeneity:** each scan draws its own realization of the shape parameters from log-normal distributions with per-parameter coefficients of variation L_{bl}: 0.04, w_{int}: 0.30, w_{out}: 0.30, A_{Ni}: 0.15, w_{Ni}: 0.14, $f_{\mathrm{Cr,bl}}$: 0.05, g_{bl}: 0.30, calibrated to the replicate-scan scatter of Veile et al.'s Fig. 9. Omitting this heterogeneity term (treating instrument noise as the only uncertainty source) produces a spuriously optimistic identifiability verdict for every parameter.
- **Verdict thresholds:** a parameter is classed identifiable if the relative standard deviation of its 20 replicate recoveries is below 0.25, weakly identifiable below 0.50, and unidentifiable otherwise. The systematic (accuracy, as opposed to precision) error of EDX quantification, of order ±10% relative, is not simulated; it floors the absolute-composition parameters $f_{\mathrm{Cr,bl}}$ and A_{Ni} independently of the statistical verdict.

S3 Nickel-exclusion moving-frame model

The nickel enrichment zone at the receding metal/barrier-layer interface (Section 3.5 of the main text) is modeled as Wagner-type solute enrichment ahead of a moving boundary, in the interface-attached frame ξ (depth into the metal ahead of the interface, $\xi \geq 0$):

$$\frac{\partial c}{\partial t} = D_{\mathrm{Ni,eff}} \frac{\partial^2 c}{\partial \xi^2} + v_m(t) \frac{\partial c}{\partial \xi}, \tag{49}$$

where the metal streams toward $\xi = 0$ at the combined interface recession speed

$$v_m(t) = \frac{1}{\mathrm{PBR_{bl}}} \frac{dL_{\mathrm{bl}}}{dt} + \phi_{\mathrm{ol,supply}} \frac{1}{\mathrm{PBR_{ol}}} \frac{dL_{\mathrm{ol}}}{dt}, \tag{50}$$

where $\mathrm{PBR_{bl}} := \Omega_{\mathrm{bl}}/\Omega_m$ and $\mathrm{PBR_{ol}} := \Omega_{\mathrm{ol}}/\Omega_m$ are the phase-specific Pilling–Bedworth ratios of the barrier and outer layers against the metal (distinct from the single unsubscripted PBR of Section S1, which refers to the barrier layer only). Numerically $\mathrm{PBR_{bl}} = 2.05$ and $\mathrm{PBR_{ol}} = 2.10$ ($FeCr_2O_4$ and Fe_3O_4 molar-volume ratios, Li et al. (2020), Table 3) and both layer growth rates are taken from the frozen zero-dimensional kinetics. The free parameter $\phi_{\mathrm{ol,supply}}$ is the fraction of the outer layer's cation supply drawn from the matrix through the barrier layer (as opposed to re-precipitation of already-dissolved species); r_{Ni}, the nickel incorporation ratio, enters through the boundary condition below.

Boundary condition (rejection flux at the moving interface, $\xi = 0$):

$$D_{\mathrm{Ni,eff}} \left.\frac{\partial c}{\partial \xi}\right|_{\xi=0} = -v_m(t)\,(1 - r_{\mathrm{Ni}})\,c(0,t), \tag{51}$$

which in the steady limit reduces to the classical rejection result $c(0) = c_m/r_{\mathrm{Ni}}$. **Far-field condition**: $c(\xi \to \infty, t) = c_m$, the bulk matrix nickel content ($c_m = 10$ wt%). **Initial condition**: $c(\xi, 0) = c_m$ everywhere (no pre-existing enrichment).

The model is solved by method-of-lines finite differencing on $\xi \in [0, 300]$ nm with 150 nodes and Radau time integration (relative tolerance 10^{-8}, absolute tolerance 10^{-10}), not by the NSEM machinery, since the three free parameters ($D_{\mathrm{Ni,eff}}$, r_{Ni}, $\phi_{\mathrm{ol,supply}}$) are fit directly to four observables (zone widths at 72/168/480 h and the 72 h peak amplitude) by nonlinear least squares. Observed features are extracted from the solved profile using Veile et al.'s own convention: zone width $W(t)$ is the extent over which $c > c_m + 2$ wt%, and amplitude $A(t) = c(0,t)$.

S3.1 Fitted values under the rejected constant-mobility model

The constant-mobility model above is rejected by the data: $\chi^2 = 25.71$ on one degree of freedom, with the discrepancy in the *shape* of $W(t)$ (predicted monotonically growing, measured approximately flat). Because parameter estimates and nested-model $\Delta\chi^2$ comparisons obtained under a structurally rejected model are not interpretable as estimates or as evidence, the fitted values are recorded here for completeness and reproducibility only, and are not used to support any claim in the main text. Table S6 gives them.

S3.2 Identifiability gate for the time-dependent mobility closures

Two one-parameter extensions were tested against the shape failure: a defect-flux-coupled effective mobility, $D_{\mathrm{Ni,eff}} \propto (\mathrm{d}L_{\mathrm{bl}}/\mathrm{d}t)^{n_{\mathrm{flux}}}$, tied to the identified growth kinetics; and a finite-capacity interfacial trapping closure with saturable trap density S_{cap}. Each was subjected to the same

Table S6: Moving-frame nickel fit to the measured zone widths and 72-h peak amplitude, *under a model the data reject in shape.* The values are recorded for reproducibility and carry no inferential weight. $D_{\mathrm{Ni,eff}}$ and r_{Ni} are jointly, not individually, constrained: both are pinned in combination (large $\Delta\chi^2$ along most directions) but trade off along one correlated direction, so neither has a one-dimensional profile interval.

Parameter	Fit	Profile 1σ	Max. $\Delta\chi^2$	Note
$D_{\mathrm{Ni,eff}}$ (nm^2/h)	3.905	correlated with r_{Ni}	101.6	jointly constrained
r_{Ni}	0.2954	correlated with $D_{\mathrm{Ni,eff}}$	123.6	jointly constrained
$\phi_{\mathrm{ol,supply}}$	0.5708	[0.579, 0.895]	3.6	weak
$\chi^2/\mathrm{dof} = 25.71/1$; with $\phi_{\mathrm{ol,supply}}$ fixed to zero: $\chi^2 = 32.1$.				

Table S7: Identifiability gate for the two candidate time-dependent Ni-mobility closures; both fail on synthetic data at this dataset's noise level and exposure-time count. RSD is the 20-replicate relative standard deviation of the recovered parameter.

Closure	New param.	Truth	RSD	Verdict
Flux-coupled	n_{flux}	1.5	27%	not identifiable (destabilizes D_{Ni} to 37%)
Finite-capacity	S_{cap}	1125	179%	not identifiable

twenty-replicate synthetic recovery gate used for the spatial shape parameters (Section S2.5) *before* being fitted to real data. Both fail (Table S7), so neither was carried to the real observables.

A forward-only scan of n_{flux} over $[-2, 3]$ at the frozen base kinetics confirms the negative result independently: no single value is simultaneously shape-correct (flat or non-monotonic $W(t)$) and a quantitative improvement on the constant-mobility χ^2.

S4 NSEM implementation and training details

This section documents every equation, loss term, boundary/initial condition, and hyperparameter of the four NSEM training configurations used in this paper, at the level of detail needed to reproduce them independently of the released code.

S4.1 Shared representation

Every configuration represents its field(s) on a discrete-variational-representation (DVR) collocation node set (Gauss–Lobatto–Legendre nodes, optionally log-clustered toward one end via a compression parameter α) using an element network with hidden width 32 and 3 layers, either a standard multilayer perceptron or a Kolmogorov–Arnold network backbone (polynomial degree 4). Thickness fields are passed through a softplus nonlinearity plus a floor of 10^{-6} to guarantee strict positivity. All training uses double-precision (float64) arithmetic.

S4.2 Configuration 1: zero-dimensional forward solve

Two time-only element networks, $\lambda_{\mathrm{bl}}(\tau)$ and $\lambda_{\mathrm{ol}}(\tau)$ (nondimensional barrier- and outer-layer thickness), on $N_t = 24$ DVR nodes over $\tau \in [0, 1]$ (log-clustered, compression $\alpha_t = 1.5$, mapping toward $\tau = 0$). Training proceeds in two stages: (1) an initial-condition pretrain (800 Adam steps at learning rate 10^{-2} plus an L-BFGS polish) that drives both networks to their constant initial-condition values, $\lambda_{\rm bl} = \lambda_0$ and $\lambda_{\rm ol} = \lambda_{\rm ol,0}$ everywhere; (2) physics training against the Crank–Nicolson residuals of the nondimensional ODE system of Section S1, with three loss terms (`ode_bl`, `ode_ol`, `ic`) combined by the BRDR aggregator (equal initial weights, adaptively rebalanced during training) and optimized for 2000 Adam steps (learning rate 10^{-3}) followed by 3 L-BFGS phases (max. 400 iterations each, strong-Wolfe line search, gradient/step tolerances $10^{-13}/10^{-15}$). The initial-condition loss carries a fixed multiplier $\lambda_{\rm ic} = 10$. Verification: relative L^∞ error against the closed-form solution, acceptance $\leq 0.5\%$.

S4.3 Configuration 2: zero-dimensional inverse, hard mode

The kinetic parameters $\{A_{\rm bl}, b_3, {\rm PBR}_{\rm eff}, L_0, L_{\rm ol,0}\}$ (optionally $C_{\rm bl}$) are the only learnable quantities, parameterized in log space (negative-log for b_3) to guarantee sign and positivity by construction. The thickness trajectories are generated by differentiable fourth-order Runge–Kutta integration of the nondimensional ODE (8 sub-steps per grid interval) through these parameters, then the affine outer-layer relation, so the governing equations are satisfied exactly and only a data-plus-prior loss is minimized:

$$\mathcal{L}_{\rm hard} = \ell_{\rm data}(\lambda_{\rm bl}; t_{\rm Cr}, \bar{L}_{\rm Cr}, \sigma_{\rm Cr})\, n_{\rm Cr} + \ell_{\rm data}(\lambda_{\rm ol}; t_{\rm Fe}, \bar{L}_{\rm Fe}, \sigma_{\rm Fe})\, n_{\rm Fe} + \lambda_{\rm prior}\Big(\tfrac{\ln L_0 - \ln(2/L_c)}{0.60}\Big)^2, \tag{52}$$

where $\ell_{\rm data}$ is the mean squared, error-normalized misfit between the network trajectory (interpolated to the data times) and the observations, $n_{\rm Cr} = n_{\rm Fe} = 3$ are the exposure-time counts (matching the main-text χ^2 normalization), and $\lambda_{\rm prior} = 1$. Optimized with 4000 Adam steps (learning rate 10^{-2}) followed by one L-BFGS phase (max. 400 iterations). This is the reference inverse mode for the zero-dimensional model; the real-data run initializes at $2\times$ the reference parameter values (`init_scale`=2.0) to verify convergence is not an artifact of a favorable initialization.

S4.4 Configuration 3: zero-dimensional inverse, soft mode

Identical field representation and pretraining to Section S4.2 (kinetics frozen during pretrain; pretrain targets are *linear ramps* from the initial condition to the last data point, not constants, so the kinetic parameters receive a nonzero time-variation gradient once joint training begins), then joint training of the field networks and the kinetic parameters against six loss terms combined by BRDR:

$$\mathcal{L}_{\rm soft} = \lambda_{\rm phys}\big(\ell_{\rm ode,bl} + \ell_{\rm ode,ol}\big) + \lambda_{\rm ic}\,\ell_{\rm ic} + \lambda_{\rm data}\big(\ell_{\rm data,bl} + \ell_{\rm data,ol}\big) + \lambda_{\rm prior}\,\ell_{\rm prior}, \tag{53}$$

with $\lambda_{\rm phys} = 1000$ (fixed multiplier on the ODE residual terms; without it the joint optimum leaves residual slack that is absorbed into the trajectory instead of the kinetics, the origin of failure mode F-1), $\lambda_{\rm ic} = 10$, $\lambda_{\rm data} = 100$, and $\lambda_{\rm prior} = 1$. The two exhibits reported in the main text (Table 6) use $\lambda_{\rm data} = 10$ as an independent reweighting check. Optimized with 4000 Adam steps (learning rate 10^{-3}) and 3 L-BFGS phases (max. 400 iterations each). The self-consistency check re-solves the recovered kinetic parameters through the exact closed-form/RK4 forward solve and compares the resulting χ^2 to the trajectory χ^2; consistency is declared satisfied if $\chi^2_{\rm exact} < \max(2\chi^2_{\rm traj}, \chi^2_{\rm traj} + 2)$. The constants in this acceptance rule are a working heuristic (tolerate up to a doubling, or an absolute slack of two χ^2 units for near-zero trajectory misfits), not a derived statistical threshold; the failure-mode exhibits in the main text exceed it by more than an order of magnitude, so the qualitative verdict does not depend on the specific constants. The specific χ^2 values of a soft-mode run vary from training run to training run (an earlier run of the $\lambda_{\rm data} = 10$ configuration landed at trajectory $\chi^2 = 0.004$ against closed-form $\chi^2 = 47.3$), but the orders-of-magnitude inconsistency signature is stable across runs; the main text quotes single-seed (seed 0) values.

S4.5 Configuration 4: spatial extension (steady and transient)

Steady (Section S2): a single field network on $N_x = 24$ DVR nodes, uniformly spaced on $\xi \in [0, 1]$ ($\alpha_x = 0$; the Péclet numbers in this problem, 10–14, produce only mild boundary layers, so uniform spacing suffices). The barrier-layer thickness is fixed at its 480-h value from the zero-dimensional fit. The loss adds the first-integral form of the steady equation, $(\hat{J} - s)^2$ averaged over all nodes, which penalizes flux non-conservation directly rather than differentiating a second-derivative residual, to the squared Robin-boundary violation at the annihilation node, trained for 2000 Adam steps (learning rate 10^{-3}) after a 500-step pretrain to the boundary value, then L-BFGS polish. The independent Newton reference solves the same first-integral system on the identical node set by Newton iteration (converges in one step, since the discrete system is linear; verified to residual $< 10^{-13}$).

Transient: field network on the tensor-product grid of the same $N_x = 24$ spatial DVR nodes and $N_t = 16$ temporal DVR nodes ($\alpha_t = 1.5$, log-clustered, $\tau \in [\tau_0, 1]$ with $\tau_0 = 0.05$). Crank–Nicolson residuals of Eq. (48) are combined with the two per-time-node boundary residuals and the initial-condition residual (four loss terms: `pde`, `bc_gen`, `bc_ann`, `ic`), BRDR-aggregated and optimized identically to the steady case.

S4.6 Hyperparameter summary

Table S8 collects every numerical hyperparameter referenced above for exact reproduction.

Table S8: NSEM hyperparameters by configuration.

Parameter	Forward	Inverse (hard)	Inverse (soft)	Spatial (steady/transient)
N_t (time nodes)	24	24	24	— / 16
N_x (space nodes)	—	—	—	24 / 24
α_t, α_x	1.5	1.5	1.5	0.0 / 1.5
Backbone	MLP/KAN	—	MLP/KAN	MLP
Hidden width, layers	32, 3	—	32, 3	32, 3
Polynomial degree (KAN)	4	—	4	4
Pretrain steps	800	—	800	500
Pretrain learning rate	10^{-2}	—	10^{-2}	10^{-2}
Adam steps	2000	4000	4000	2000
Adam learning rate	10^{-3}	10^{-2}	10^{-3}	10^{-3}
L-BFGS phases × max. iter.	3 × 400	1 × 400	3 × 400	3 × 400
L-BFGS tolerances (grad, change)	10^{-13}, 10^{-15}	10^{-13}, 10^{-15}	10^{-13}, 10^{-15}	10^{-13}, 10^{-15}
λ_{ic}	10	—	10	10
λ_{data}	—	(built into ℓ_{data})	100 (or 10)	—
λ_{phys}	—	—	1000	—
λ_{prior}	—	1	1	—
Random seed	0	0	0	0

S5 Baseline numerical verification

Figure S1 summarizes the baseline verification of the forward model. The closed-form solutions, Eqs. (17) and (27), agree with a stiff Radau integration of the apparent-parameter ODE system to better than 6×10^{-11} nm on the demonstration parameter set (maximum absolute deviation 4.63×10^{-11} nm for the barrier layer and 5.07×10^{-11} nm for the outer layer), i.e. machine-level parity. In addition, the implementation reproduces the published thickness-versus-time behavior when driven by the Table-5 apparent parameters of Li et al. (2020) for HCM12A and 316L at 500°C in supercritical water, confirming that the adapted model regresses exactly to the supercritical-water formulation in its original regime.

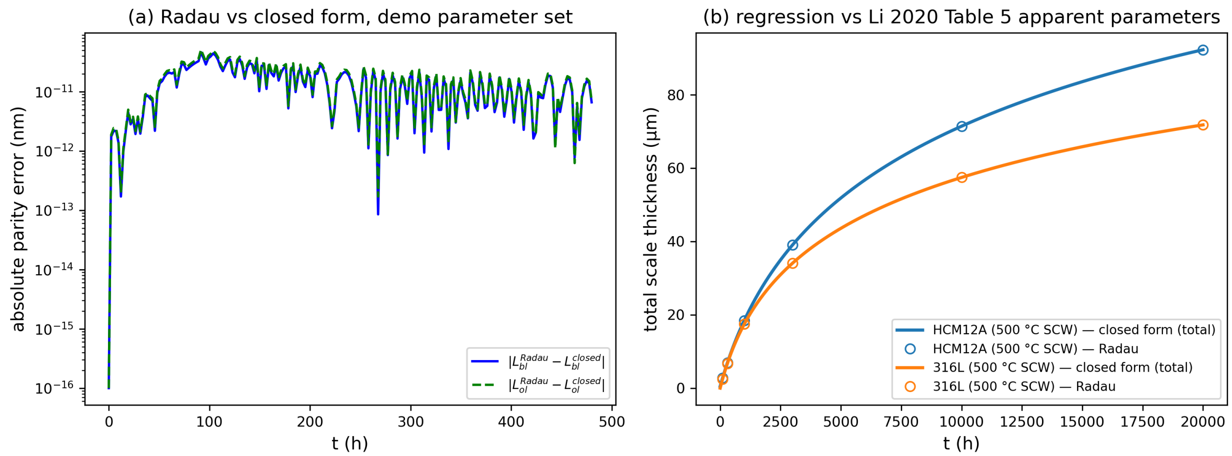

Figure S1: Baseline numerical verification. Closed-form solutions versus Radau integration for the barrier and outer layers (parity better than 6×10^{-11} nm), and regression against the Table-5 apparent parameters of Li et al. (2020) for HCM12A and 316L at 500°C in supercritical water.

S6 Quasi-steady approximation error

The interface-reaction-controlled model of Section S1 treats the defect concentration profiles inside the barrier layer as quasi-steady. Figure S2 quantifies the error incurred by this assumption using a full moving-boundary transport solve. At the end of the 480 h exposure window, the oxygen-vacancy species (whose migration transit time across the film is approximately 5 h) lags its quasi-steady profile shape by only 0.69%, while the slower cation-vacancy species (transit time approximately 36 h) lags by 5.79%. The quasi-steady closure is therefore an excellent approximation for the species that controls film growth, and an acceptable one for the slowest defect over the experimental window.

S7 Operating-envelope contour maps

Figure 10 of the main text presents the operating-envelope sensitivity as line sweeps, which is the form in which the two findings are legible: temperature carries the sensitivity, and it grows with the scanned effective activation energy, while three decades of dissolved oxygen move the prediction by under six percent at 480 h and under two percent at ten years. Figure S3 gives the same calculation as contour maps over the full $(T, [O_2])$ box, for readers who want the joint dependence rather than the two sections through it. The near-vertical contours are the same result seen a second way: at fixed temperature, the whole oxygen range spans less than one contour interval.

S8 Neural-backbone parity

The inverse-identification results in the main text are independent of the choice of neural backbone. Trained on the forward problem and compared against the closed-form solutions of Section S1, the multilayer perceptron (MLP) backbone achieves relative L^∞ errors of 0.14% for the barrier layer

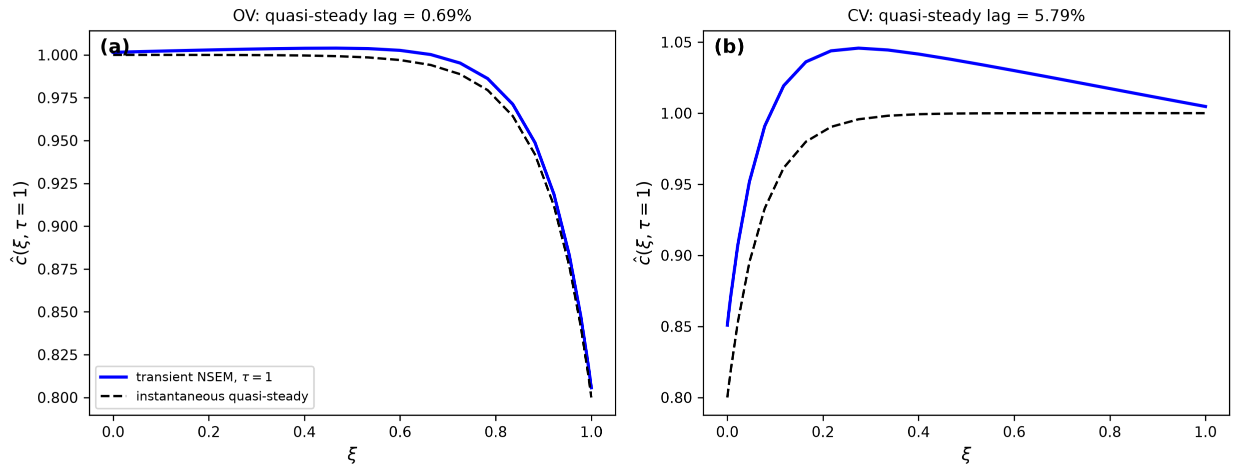

Figure S2: Quasi-steady approximation error from the moving-boundary transient solve. At $t = 480$ h the oxygen-vacancy profile (migration transit time ≈ 5 h) deviates from its quasi-steady shape by 0.69%; the cation-vacancy profile (≈ 36 h) by 5.79%.

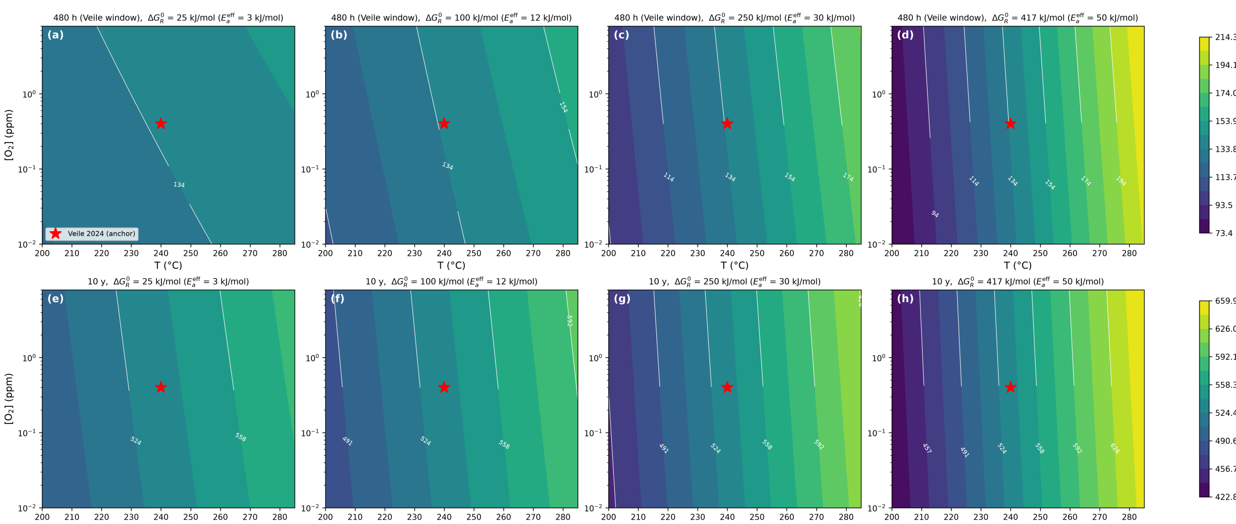

Figure S3: Operating-envelope sensitivity maps: predicted L_{bl} at (a–d) 480 h and (e–h) ten years over the scanned envelope (200–285°C, 0.01–8 ppm O_2), for $\Delta G_R^0 = 25$, 100, 250 and 417 kJ/mol, that is effective activation energies $\alpha_3 \Delta G_R^0$ of 3, 12, 30 and 50 kJ/mol. The marker locates the Veile et al. calibration condition. Each row shares one color scale, so the widening spread from left to right is the sensitivity increasing with activation energy.

and 0.12% for the outer layer, while the Kolmogorov–Arnold backbone achieves 0.11% and 0.04%, respectively. Both backbones are well inside the 0.5% acceptance bar, so the parameter estimates and uncertainty statements reported in the main text do not depend on the backbone architecture.

S9 Inversion workflow

Fig. S4 summarizes how the reduced model is inverted. It restates in one picture the procedure described in Section 2.4 of the main text and specified in full in Section S4.

S10 Bootstrap ensembles

The 1000-member parametric bootstrap summarized in Table 4 of the main text is shown here parameter by parameter against the profile-likelihood band, which is where the two weakly constrained parameters fail in opposite directions.

S11 The F-1 failure mode of soft-mode inversion

Section 3.3 of the main text reports a failure mode specific to sparse-data physics-informed inversion: a soft-mode solve whose trajectory passes through every measurement while the parameters it reports do not satisfy the governing equation. The exhibit is reproduced here.

Table S9: F-1 self-consistency exhibit: trajectory χ^2 versus closed-form χ^2 at the recovered parameters (seed-0 runs; see Section S4 for run-to-run variability).

Inverse mode	Note	χ^2 traj.	χ^2 closed-form	Self-consistent
Hard (real data)	physics exact by construction	5.793	5.79	yes
Soft, $\lambda_{\text{phys}} = 10^3$	joint field+parameter	0.21	16.64	no
Soft, $\lambda_{\text{data}} = 10$	joint field+parameter	0.67	15.57	no

S12 Synthetic identifiability gate for the spatial shape parameters

Before the spatial extension of Section 3.4 of the main text is fitted to the measured composition maps, its shape parameters are put through a synthetic recovery gate under the instrument and heterogeneity model of Section S2.5.

S13 Steady spatial solve

The steady defect-transport solve underlying Section 3.4 of the main text, verified against the analytic solution of Section S2 and against an independent Newton solve on the same node set.

S14 Operating-envelope line sweeps

Table 6 of the main text gives the envelope spreads; the sweeps behind them are shown here, with temperature and dissolved oxygen varied one at a time on a common vertical scale.

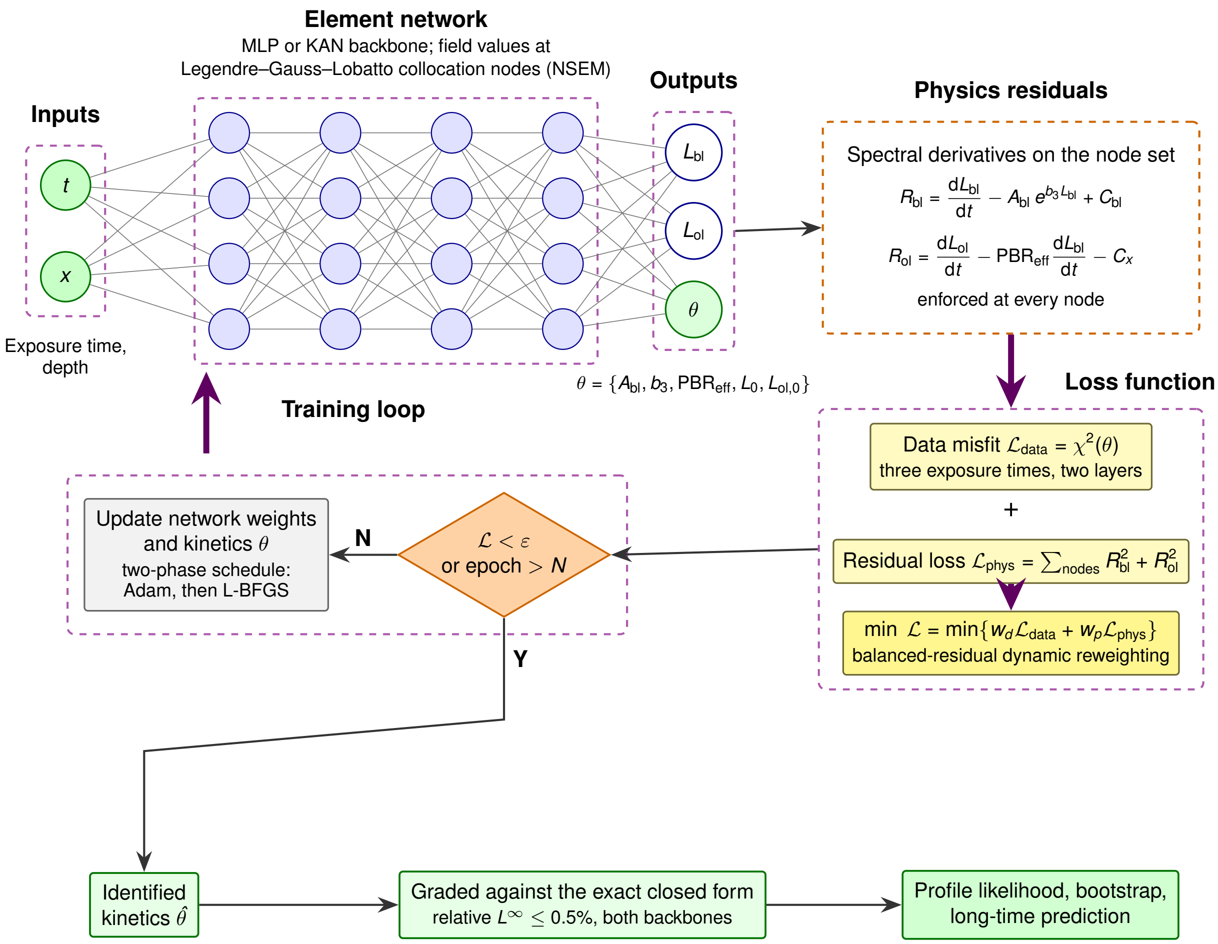

Figure S4: How the reduced model is inverted. Exposure time, and for the spatial extension depth, enters an element network whose field values are evaluated at Legendre–Gauss–Lobatto collocation nodes; the network returns the two layer thicknesses and the five kinetic parameters θ. Spectral differentiation on the same node set gives the residuals of Eqs. (10)–(11) of the main text, which are enforced at every node and combined with the data misfit of Eq. (16) of the main text by a balanced-residual dynamic reweighting of the two terms; a two-phase schedule, Adam then L-BFGS, drives the total loss to convergence. The identified kinetics are graded against the exact closed form of Eqs. (13)–(14) of the main text before they are used for profile likelihood, the bootstrap, and the long-time prediction. The solver is the neural spectral-element method (NSEM) of Tetsassi Feugmo and Pankaczy; architecture and hyperparameters for each configuration are tabulated in Section S4.

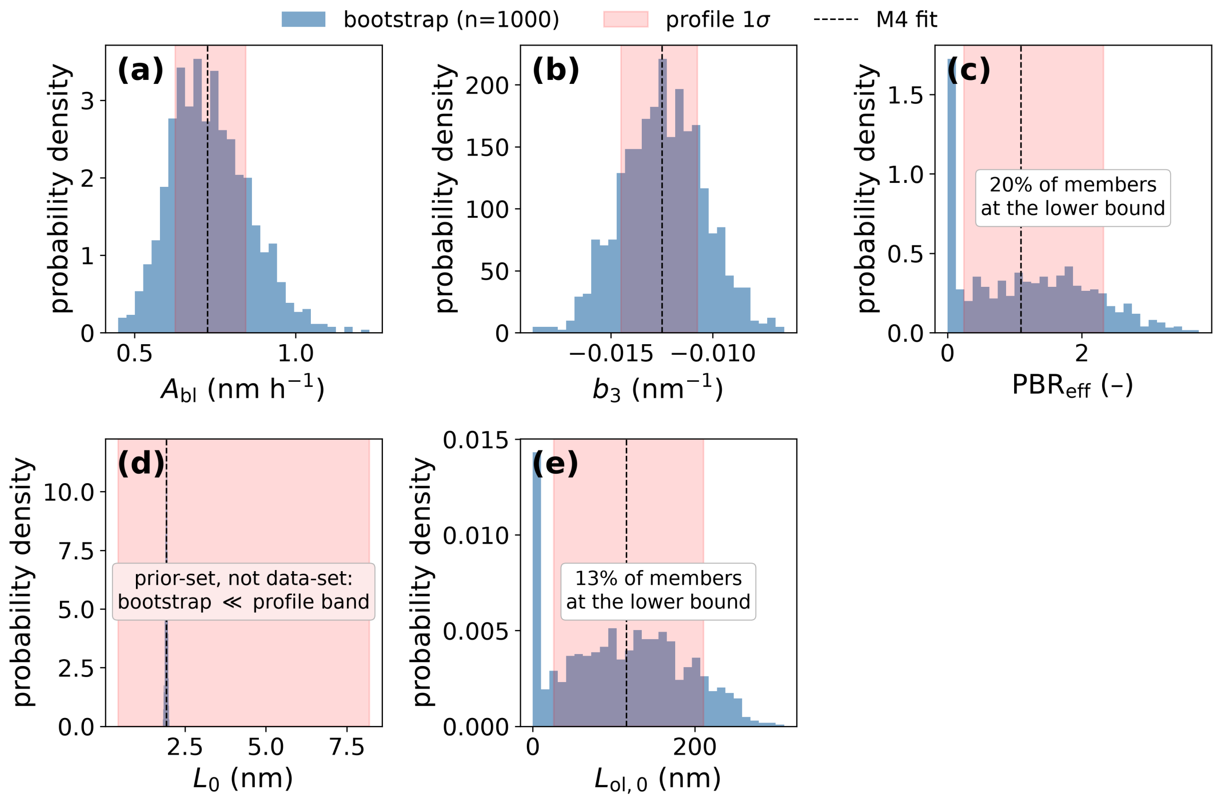

Figure S5: Bootstrap ensemble versus profile likelihood: 1000-member parametric bootstrap distribution for each M4 parameter, compared against the profile-likelihood 1σ band. Panels follow the order of Table 4 of the main text: (a) A_{bl}, (b) b_3, (c) $\mathrm{PBR}_{\mathrm{eff}}$, (d) L_0, (e) $L_{\mathrm{ol},0}$. In each, the blue histogram is the distribution of the 1000 refitted values (plotted as a probability density, so panel heights are not comparable across parameters), the red shaded band is the profile-likelihood 1σ interval of Table 4 of the main text, and the dashed vertical line is the M4 point estimate. Two annotated features are results rather than plotting artifacts. The spike at the lower bound in (c) and (e) is the fraction of resampled datasets for which the parameter is driven to zero, which is what weak identifiability looks like in a resampling ensemble. In (d) the bootstrap is far narrower than the profile band: resampling the data cannot explore a direction the data do not constrain, so the narrow histogram is not a tighter constraint but the signature of a prior-set parameter.

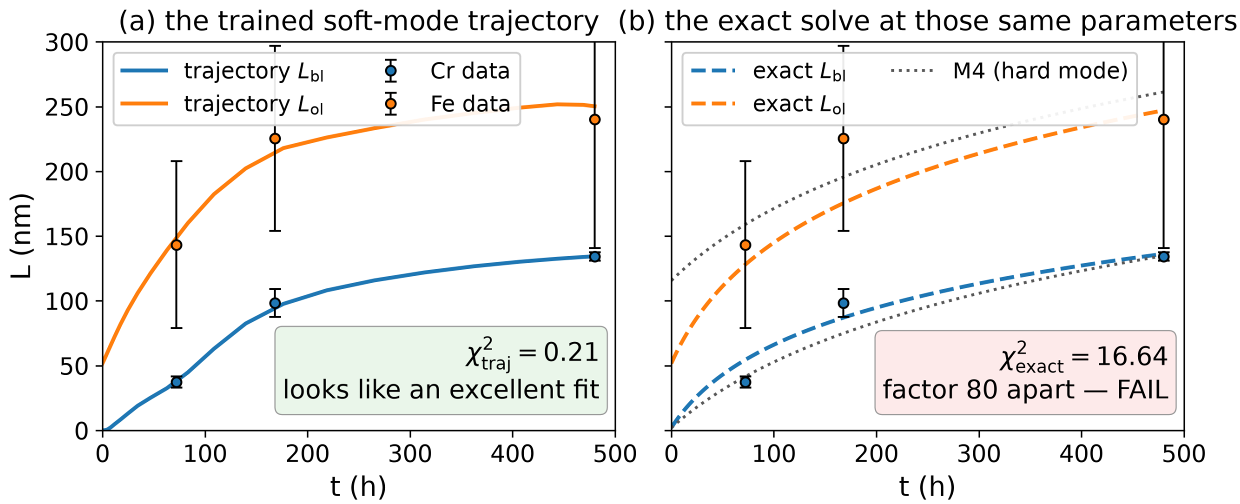

Figure S6: The F-1 failure mode. Both panels use the same six measurements and the same recovered parameters; in both, blue denotes the barrier layer and orange the outer layer, and the points are the measured means with their standard errors. (a) The trained soft-mode trajectory (solid curves), which is what the data loss sees: it passes through every point, at a χ^2 of 0.21 that is lower than the accepted model's 5.79. (b) The exact forward solution (dashed curves) evaluated at exactly the parameters that trajectory reports, against the same data, with the hard-mode M4 solution (gray dotted) for reference: $\chi^2 = 16.6$. The trajectory absorbed the misfit as slack in the governing equation, so the apparent quality of the fit in (a) is not a property of the recovered parameters. The factor of roughly 80 between the two is the diagnostic; the hard mode returns 5.79 by either route and passes by construction.

Table S10: Synthetic identifiability gate for the spatial shape parameters: 20-replicate recovery under the instrument and heterogeneity model of Section S2.5. RSD is the replicate relative standard deviation of the recovered parameter; the verdict thresholds are 25% (identifiable) and 50% (weak). The L_{bl} truth value is a round 134.0 nm, set from the 480-h kinetic-fit output of Table 3 of the main text (134.8 nm).

Shape param.	Physics mapping	Truth	Unit	RSD	Verdict
L_{bl}	kinetic-fit output	134.0	nm	2%	identifiable
w_{int}	solid-state interdiffusion	5.0	nm	17%	identifiable
w_{out}	outer-interface kinetics	8.0	nm	19%	identifiable
A_{Ni}	Ni exclusion ratio	25.0	wt%	7%	identifiable
w_{Ni}	Ni exclusion + transport	35.0	nm	8%	identifiable
$f_{Cr,bl}$	interfacial flux ratios	45.0	wt%	2%	identifiable
g_{bl}	defect-transport gradient	1.0	wt%	47%	weak

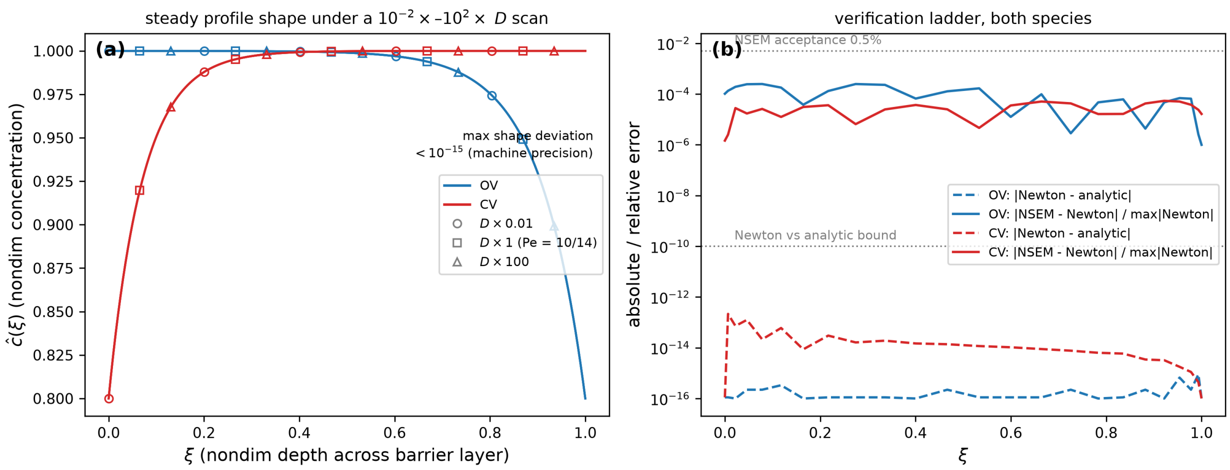

Figure S7: Steady spatial solve. (a) Steady-state nondimensional profile shape (nondimensional concentration $\hat{c}$ against nondimensional depth ξ across the barrier layer) with the diffusivity scaled by $0.01\times$ to $100\times$. Blue curves are oxygen vacancies (OV) and red curves cation vacancies (CV); the three diffusivity scalings are distinguished by marker symbol and coincide to machine precision, which is the diffusivity-cancellation result. (b) Verification ladder for the same two species (same colors): dashed curves are the absolute Newton-versus-analytic error and solid curves the NSEM-versus-Newton error relative to the peak Newton value, both against depth on a logarithmic ordinate; the dotted horizontal lines mark the 10^{-10} and 0.5% acceptance bounds.

S15 The identified kinetics in context

Table S11: The identified kinetics in the context of related systems. Literature thickness entries are indicative orders of magnitude for the Cr-rich inner layer; field strengths are apparent values as reported by the cited analyses.

System	Environment	T (°C)	Inner layer	ε_f (V/cm)
AISI 347 (this work)	simulated BWR HTW	240	41–135 nm (72–480 h)	4×10^{3}–2×10^{4}
304/316 SS	oxygenated HTW (Kuang et al. 2010; Kim 1999)	288	tens–hundreds nm	not reported
HCM12A, 316L	supercritical water (Li et al. 2020)	500	μm scale	$\sim 10^{2}$
Passive metals	ambient aqueous (Cabrera and Mott 1949)	25	1–5 nm	$\sim 10^{6}$

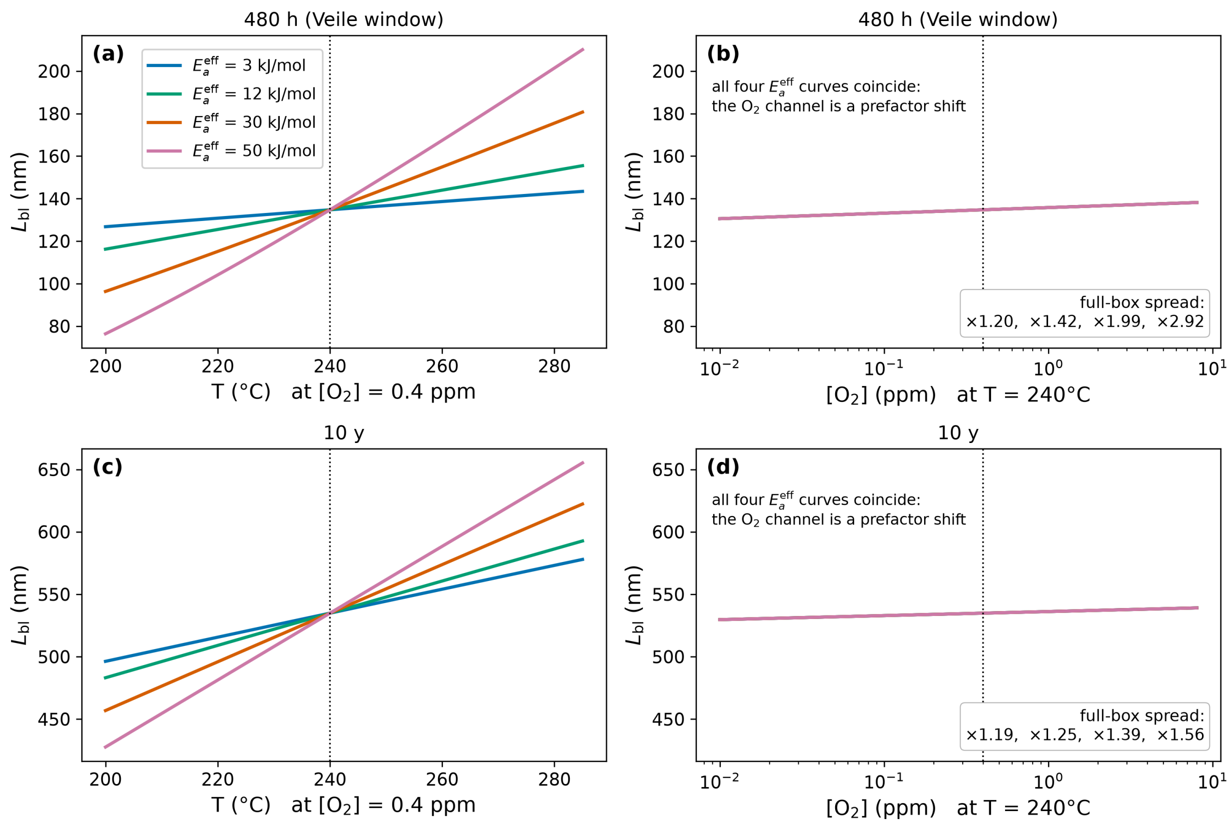

Figure S8: Operating-envelope sensitivity of the predicted barrier-layer thickness, at (a, b) 480 h and (c, d) ten years. Left panels sweep temperature at the calibration oxygen level; right panels sweep dissolved oxygen across three decades at the calibration temperature, on the same vertical scale. Each curve is one scanned value of the effective activation energy $\alpha_3 \Delta G_R^0$, spanning 3–50 kJ/mol; the dotted line marks the Veile et al. calibration condition, through which every curve passes by construction. Temperature carries the sensitivity and it grows with activation energy, while three decades of dissolved oxygen shift the prediction by under six percent at 480 h and under two percent at ten years, and do so identically at every activation energy, the oxygen channel being a prefactor shift. Boxed factors are the full-envelope spreads of the four plotted activation energies, from Table 6 of the main text. Contour maps over the full $(T, [O_2])$ box are given in Fig. S3.